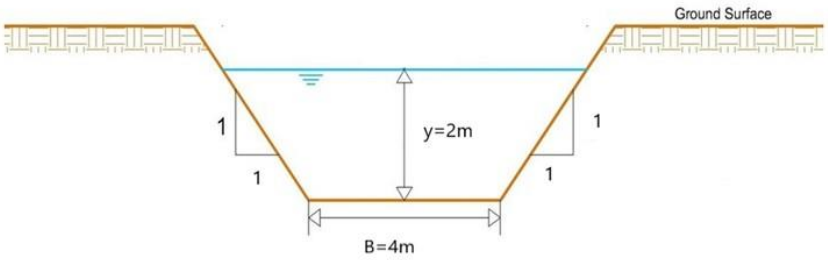
2020-2021 FALL SEMESTER HYDRAULICS COURSE 3rd SHORT EXAM Date:24.12.2020

Name: **Surname:** **Student ID:**

Underline the last two digits of your student number you wrote above. The last two digits of your student ID number will be the "**ab**" value.

QUESTION (25P)

The trapezoidal channel given in the figure has a bottom slope of J_0=0.0025. The roughness coefficients are 0.014 and 0.025 for the side slopes and the bottom, respectively.



a) Determine the flowrate of the channel.
b) Determine the critical flow conditions (critical depth, critical velocity, critical slope and minimum energy) for this flowrate.
c) Determine the flow regime.
d) For an energy "**E_{min}+(1.ab)**" metre, what will be the uniform flow depths? Draw the specific energy diagram for this flowrate.
e) Determine the depth of water and the flowrate for the ***most economical hydraulic section*** in a trapezoidal channel (B = 4m) with side slopes of 1/1.5 and a roughness of 0.025 on all surfaces (J_0=0.0025).